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Cooperative Mission Planning for Heterogeneous Robots with Energy Constraints

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Introduction

Multi-robot systems have gained significant attention for their ability to handle complex tasks in various applications. A key challenge in these systems is the **Multi-Robot Task Assignment** (**MRTA**) problem. Addressing MRTA becomes more complex in scenarios involving energy constraints and dynamic, unknown environments.

This research explores a **cooperative mission planning problem** where a master vehicle serves as a moving depot for agent vehicles. The master vehicle follows a predefined path while agents execute tasks and return for recharging. This approach is particularly suited for environments where deploying static depots is impractical.

Our work stands out by enabling **task assignment and mission planning in dynamic environments** where tasks are detected in real time. This allows the system to adapt to new mission requirements efficiently while addressing **energy limitations** of agents.

Problem description

System elements:

- A ground vehicle with motion model $\dot{\boldsymbol{p}}_r(t) = f(\boldsymbol{p}_r(t), \boldsymbol{v}_r(t), \boldsymbol{a}_r(t))$, a maximum velocity $\boldsymbol{v}_{r,max}$(m/s) and equipped with a sensor to sense tasks information.
- A fleet of drones with battery capacity for each b_k and the average speed $\boldsymbol{v}_d$ (m/s)
- The set of available tasks $Q(t)$

Mathematical model:

- Objective function:

$$maximize \quad \sum_{j=1}^{m} |R_j|$$

$$s.t. \quad d_{\boldsymbol{p}_r(t),r_1} + \sum_{i=2}^{l-1} d_{r_i,r_{i+1}} + d_{r_l,\boldsymbol{p}_r(t')} \leq b_k|\boldsymbol{v}_d|$$

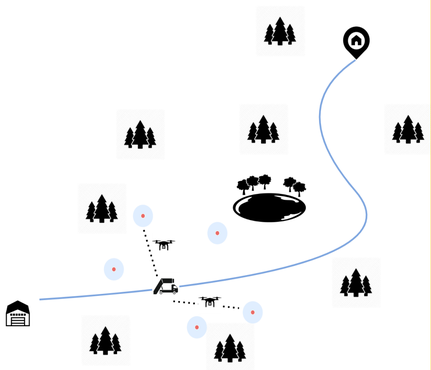


Fig.1: System description

Methodology

The task allocation is a sequential process that consist 4 key steps:

1. Priority calculation

Decide which tasks should be visit first

$$\rho_{q_i}(t) = -\frac{\boldsymbol{v}_r(t)}{|\boldsymbol{v}_r(t)|}(\boldsymbol{q}_i - \boldsymbol{p}_r(t))^T$$

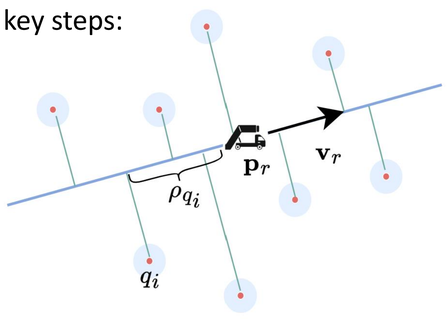


Fig.2: Prioritization of tasks

3. Route elimination

Eliminate routes that are not ready to improve system efficiency by checking the condition:

$$\sum_{r_i \in R_j} \|r_i - \boldsymbol{p}_r(t)\| \left(4 + 2\left(\frac{|\boldsymbol{v}_{r,max}|}{|\boldsymbol{v}_d - \boldsymbol{v}_{r,max}|}\right)\right) \geq b_k|\boldsymbol{v}_d|$$

4. Route length optimization

We use 2-opt TSP [1] for optimize traveling distance of drones

Additional

We adjust the velocity of the carrier based on current available tasks

$$A = \frac{|\{q_i|\rho_{q_i} > 0 \wedge q_i \in Q(t)\}| - |\{q_i|\rho_{q_i} \leq 0 \wedge q_i \in Q(t)\}|}{|Q(t)|}$$

$$|\boldsymbol{v}_r(t)| = |\boldsymbol{v}_r(t)| - A|\boldsymbol{a}_r(t)|$$

2. Route planning

Algorithm 1: Route planning for a drone

```
Result: A feasible route R
B ← b_k ;
q_p ← p_r(t) ;
R ← [] ;
while B > 0 and not Q.empty do
    nextNode ← findBestNext() ;
    if nextNode then
        timeTravel ← distance(q_p, nextNode) / |v_d| ;
        R.append(nextNode) ;
        B ← B - timeTravel ;
        q_p ← nextNode ;
        Q.delete(nextNode) ;
    else
        break ;
```

where the $findBestNext()$ function is with:

$$maximize \quad \rho_{q_i}$$

$$s.t. \quad d_{q_p,q_i} + d_{q_i,\boldsymbol{p}_r(t')} < \mathcal{B}$$

with $\mathcal{B}$ is the remaining battery when drone finish task q_p

the value of $\boldsymbol{p}_r(t') = \boldsymbol{p}_r(t_p + t_1 + t_2)$ can be bounded with

$$maximize \quad t_2$$

$$s.t. \quad z + d_{(x,y,0),\boldsymbol{p}_r(t')} \leq |\boldsymbol{v}_d|(\mathcal{B} - t_1)$$

$$t_2 \leq (\mathcal{B} - t_1)$$

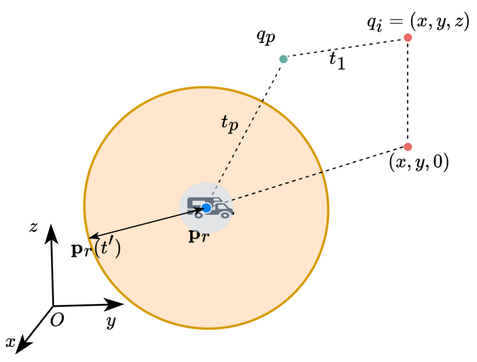
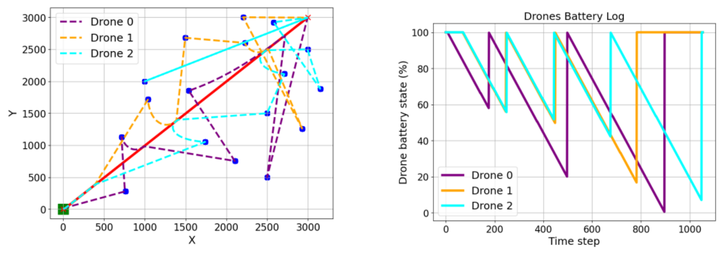


Fig.3: Illustration in bounding $\boldsymbol{p}_r(t')$

Algorithm 2: The general planning system

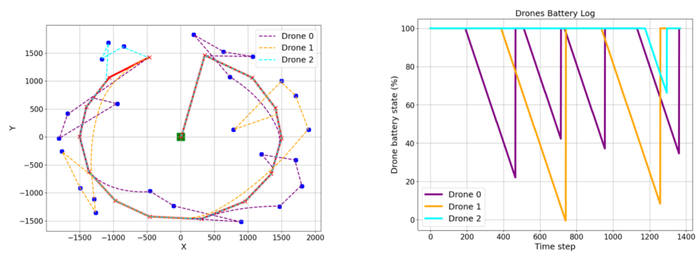
```
while not carrier_robot.at_goal() do
    carrier_robot.calculate_priority() ;
    control_input = carrier_robot.control_by_priority() ;
    carrier_robot.move(control_input);
    for k in droneIndex do
        if drone_k.state() == working then
            drone_k.move_to(R_assigned_for_drone_k);
        if drone_k.state() == charging then
            continue;
        if drone_k.state() == waiting then
            R_new = carrier_robot.plan() ;
            if R_new.is_ready() ;
            then
                R_new.optimize() ;
                drone_k.assign(R_new);
```

Result



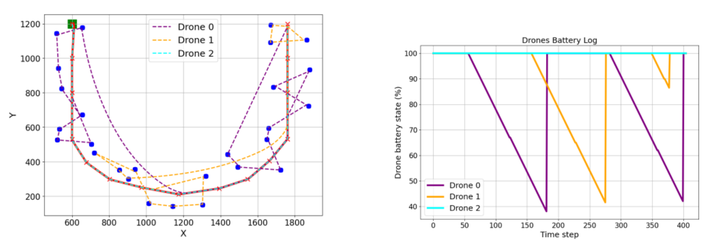
(a) 2D illustration of drone paths (b) Battery states of drones

Fig. 5: Straight path with drone speed 12 m/s, drone battery limit 15 (min)



(a) 2D illustration of drone paths (b) Battery states of drones

Fig. 6: Circle path with drone speed 15 m/s, drone battery limit 20 (min)



(a) 2D illustration of drone paths (b) Battery states of drones

Fig. 7: U-curve path with drone speed 20 m/s, drone battery limit 30 (min)

Table 2: Simulation result

	Straight path			Circle path			U-curve path		
Drone maximum speed	12	15	20	12	15	20	12	15	20
Drone battery limit	15	20	30	15	20	30	15	20	30
Number of assigned tasks	17	17	17	25	25	25	27	27	27
Number of routes	10	7	5	12	7	6	7	5	4
Number of successful drones	3	3	3	3	3	3	3	3	3

Conclusion and Future research

In this paper, we present a new approach to the problem of multi-robot task allocation in an unknown environment, extending the assumption that agent vehicles can return during the movement of the master vehicle. By optimizing the upper bound of travel time to meet energy constraints, we successfully generate a safety plan for the fleet. However, one limitation of our solver is that it does not ensure an equal task distribution among agents. We aim to address this issue in future research. Additionally, an online strategy for assigning new tasks to agents during their operation to further optimize the plan can be achieved by the model of the distributed system.

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References

[1] Englert, M., Röglin, H., Vöcking, B.: Worst case and probabilistic analysis of the2-opt algorithm for the tsp. Algorithmica (2014)

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